

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

HTTP

- **Full Form:** Hypertext Transfer Protocol
- **Use:** It is the foundational protocol used to transmit web pages and data across the internet from a server to your browser.

HTTPS

- **Full Form:** Hypertext Transfer Protocol Secure
- **Use:** It is the secure version of HTTP that encrypts data transmissions to protect sensitive information (like passwords or credit cards) from being intercepted.

FTP

- **Full Form:** File Transfer Protocol
- **Use:** It is a dedicated protocol used specifically for uploading, downloading, and transferring files between a client and a server on a network.

SMTP

- **Full Form:** Simple Mail Transfer Protocol
- **Use:** It is the standard protocol used by email clients and servers to send and relay outgoing emails across the internet.

2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

IPv4 uses a 32-bit address space to provide roughly 4.3 billion unique addresses (e.g., 192.168.1.1), which are now largely exhausted. In contrast, IPv6 uses a 128-bit address space to offer a virtually infinite number of unique addresses (e.g., 2001:db8::ff00:42:8329) alongside built-in security and more efficient routing.

3. Run nslookup google.com on your system and write down the IP address you get.

Microsoft Windows [Version 10.0.26200.8457]

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C:\Users\RIDHAM>nslookup google.com

Server: UnKnown

Address: 10.58.31.104

Non-authoritative answer:

Name: google.com

Addresses: 2404:6800:4002:80b::200e

172.217.31.14

Google.com

Addresses: 2404:6800:4002:80b::200e

172.217.31.14

4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

200 OK

- **Meaning: Success.** The request was successful, and the server has delivered the data or webpage you asked for.

301 Moved Permanently

- **Meaning: Permanent Redirect.** The URL of the requested resource has been changed permanently. The server automatically routes you to the new address.

403 Forbidden

- **Meaning: Client Error (No Permission).** The server understands your request, but you do not have the access rights or permissions to view the content (e.g., trying to access an admin panel without being logged in).

404 Not Found

- **Meaning: Client Error (Missing Page).** The server cannot find the requested webpage or file. This usually happens if the link is broken, the page was deleted, or there was a typo in the URL.

500 Internal Server Error

- **Meaning: Server Error.** A generic error message indicating that something went wrong on the website's server side (like a crash or a bug in the code), preventing it from fulfilling your request.

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

1. The Root Zone (.)

- **What it is:** The absolute top of the DNS hierarchy. It is represented by a silent dot (.) at the very end of a domain name. While browsers automatically hide this trailing dot today, it is explicitly parsed by the internet's core infrastructure.
- **In our example:** The invisible dot at the end: www.google.com.

- **How it works:** When you look up a site, your computer first asks a Root Nameserver, *"Where can I find the servers that handle .com?"*

2. Top-Level Domain - TLD (.com)

- **What it is:** The next level down the tree. TLDs categorize websites by purpose, organization, or geography. They are managed by specific registries (like Verisign for .com).
- **Types of TLDs:** * Generic TLDs (gTLDs): .com, .org, .net, .edu
 - Country-Code TLDs (ccTLDs): .in (India), .uk (United Kingdom)
- **In our example: .com**
- **How it works:** The Root server points your request to the .com TLD nameserver. Your computer then asks this TLD server, *"Where can I find the specific server for google?"*

3. Second-Level Domain - SLD (google)

- **What it is:** This is the unique name a person or company registers with a domain registrar (like GoDaddy or Namecheap). It represents the specific identity or brand on the web.
- **In our example: google**
- **How it works:** The TLD server points your request to Google's Authoritative Nameservers. This is the server that holds the master record of exactly which IP addresses belong to Google's infrastructure.

4. Subdomain (www)

- **What it is:** A folder or subsection created *under* the primary domain. Domain owners can create unlimited subdomains to organize different services or servers without buying a new domain name.
- **In our example: www** (World Wide Web)
- **Other variations:** You could also have **mail**.google.com or **drive**.google.com.
- **How it works:** Google's authoritative nameserver reads the **www** part and says, *"Ah, you want our main web server. Here is the exact IP address you need."*

